

Long-term outcome in patients with Guillain-Barré syndrome requiring mechanical ventilation

Dade D. Fletcher, MD; Nicholas D. Lawn, FRACP; Troy D. Wolter, MS; and Eelco F.M. Wijdicks, MD

Article abstract—*Objective:* To analyze long-term recovery and predictors of outcome in patients with Guillain-Barré syndrome (GBS) requiring mechanical ventilation. *Methods:* The clinical and electrophysiologic data of 114 patients with GBS admitted to the intensive care unit between 1976 and 1996 (60 mechanically ventilated, 54 nonventilated) were reviewed. Functional disability and predictors of outcome were determined at 1 year and at maximal recovery using the Hughes scale. Good outcome was defined as ability to ambulate without assistance; poor outcome was defined as inability to ambulate independently. *Results:* Mechanical ventilation was required in 81% of patients with a poor outcome. Mortality was 20% in patients ventilated for GBS. However, ventilated patients who survived did well, with 79% eventually regaining independent ambulation. Nineteen percent of patients improved at least one functional grade beyond 1 year. Univariate predictors of poor maximal recovery in ventilated GBS patients were increased age ($p < 0.001$), upper limb paralysis ($p = 0.004$), duration of ventilation ($p = 0.006$), and delay of more than 2 days to transfer to a tertiary center ($p < 0.001$). However, only age (OR 1.99, $p = 0.004$) and delayed transfer (OR 19.8, $p = 0.002$) were independently predictive of poor outcome on multivariate analysis. *Conclusion:* Mechanically ventilated patients constitute the majority of GBS patients with a poor outcome, and mortality remains substantial in this subgroup (20%). Although recovery from severe GBS may be prolonged, most survivors regain independent ambulation. **Key words:** Guillain-Barré syndrome—Outcome—Mechanical ventilation.

NEUROLOGY 2000;54:2311–2315

The prognosis for recovery in patients with Guillain-Barré syndrome (GBS) is generally favorable, with the majority experiencing no or only minor residual symptoms. However, the mortality rate ranges from 2 to 12% despite technological advances in supportive care, and 15% of GBS patients have persistent disability.^{1–5} Mechanical ventilation is required in 20 to 30% of patients with GBS and is a risk factor for a poor outcome.^{3,6–8} Few multicenter studies have systematically analyzed ventilated patients and single center studies involve comparatively few patients. Follow-up has often been limited to 6 months or 1 year, which may result in an overly pessimistic outlook on long-term recovery from severe GBS. In contrast, analysis of a more homogeneous population, such as in a treatment trial, where patients with comorbidity are excluded may overestimate the proportion who make a good recovery from severe GBS.^{1,9,10} Analysis of an unselected group of ventilated patients with GBS, who are known to be at increased risk of a poor outcome, may permit determination of the most important factors predictive of outcome and a more accurate depiction of long-term recovery.

Methods. The medical records of all patients with GBS admitted to the intensive care unit (ICU) at Mayo Clinic–

affiliated hospitals between 1976 and 1996 were reviewed. All patients were examined by a neurologist. Standard diagnostic criteria for GBS were used.^{2,11} Children (younger than 16 years) and patients with atypical variants were included.²

The clinical and electrophysiologic features analyzed in the ventilated patients included the following: bulbar dysfunction (presence of impaired gag, dysarthria, or clinical history of dysphagia); autonomic dysfunction (unexplained cardiac dysrhythmia, blood pressure fluctuations, significant bowel or bladder involvement); preceding gastrointestinal (GI) illness (history of abdominal pain or diarrhea); bilateral facial palsy; time to peak disability (from onset of neuropathic symptoms); low compound muscle action potential (CMAP) amplitude ($<20\%$ lower limit of normal [LLN] in at least two nerves); inexcitable nerves (inexcitable nerves or CMAP $<10\%$ LLN in one nerve and absent in all other nerves tested); treatment (plasma exchange [PE] or IV immunoglobulin [IVIg]); and time to transfer (initially admitted to a facility elsewhere and subsequently transferred).¹² Upper limb function was also analyzed at peak disability using the following scale: 0, normal; 1, weak, but able to lift arms off bed; 2, flicker of movement; 3, paralysis (no movement).

Outcome at 1 year and at maximal recovery were determined. Patient function was ranked, according to the scale adopted by Hughes, as follows: 0, healthy; 1, minor symptoms or signs, able to run; 2, able to walk >5 m without

From the Departments of Physical Medicine and Rehabilitation (Dr. Fletcher) and Neurology (Drs. Lawn and Wijdicks) and Section of Biostatistics (T.D. Wolter), Mayo Clinic and Foundation, Rochester, MN.

Received December 13, 1999. Accepted in final form March 13, 2000.

Address correspondence and reprint requests to Dr. Wijdicks, Department of Neurology, W8A, Mayo Clinic, 200 First Street SW, Rochester, MN 55902; e-mail: wjde@mayo.edu

assistance but unable to run; 3, able to walk >5 m with assistance; 4, bed- or chair-bound; 5, requiring assisted ventilation for at least part of the day; 6, dead.^{1,13} Maximal recovery (long-term outcome) was defined as the peak functional grade achieved after which no further clinical improvement occurred, in the absence of intervening comorbidity. Patients were followed until they regained the ability to ambulate independently (grade 2). A semi-structured phone interview of patients or family members was conducted in those patients who had not reached grade 2 at the last documented follow-up. Good outcome was defined as the ability to ambulate independently (grade ≤ 2); poor outcome was defined as inability to ambulate independently (grades 3 through 6). Death as a direct result of GBS was considered if there was a relevant event in association with ongoing symptoms of GBS, or when death occurred in the context of prolonged immobility and respiratory dysfunction related to the disorder. The circumstances surrounding death in this series of patients have been described elsewhere.⁸

Statistical analysis. The clinical and electrophysiologic features of ventilated patients were analyzed according to outcome. Univariate factors predictive of maximal recovery outcome were assessed using Wilcoxon rank sum test for continuous variables and Fisher's exact test for categorical variables. Multivariate logistic regression analysis, using a backwards elimination procedure, was used to determine independent predictors of poor maximal recovery in the ventilated GBS patients. Maximal recovery (good versus poor) was the dependent variable and all variables found to be significant on univariate analysis were included in the model as independent variables. All significant variables were entered into the model at the first step. The most nonsignificant variable was then removed, with all remaining variables entered into the model at the second step. This procedure was repeated until the variables remaining were those that contributed significantly ($p < 0.05$) to the prediction of outcome. Separate analyses were carried out for the 1-year outcome to allow comparison with previous studies. Odds ratios (OR) and corresponding 95% confidence intervals for variables found on multivariate analysis to be predictive of poor outcome were calculated. In all cases, two-tailed p values ≤ 0.05 were considered as evidence of findings not attributable to chance.

Results. A total of 114 ICU patients with GBS (grade ≥ 3) were identified, of whom 60 required mechanical ventilation. This included 10 children (six ventilated) and six patients who presented initially with features of Miller Fisher variants (two ventilated). Reasons for admission to the ICU included rapidly progressive weakness, concern regarding impending respiratory failure, or dysautonomia. Condition at 1 year and at maximal recovery were available in 54 of the 60 ventilated patients (90%) and in 51 of the 54 nonventilated patients (94%). Follow-up using a semi-structured phone interview was performed in 12 ventilated patients and in 4 nonventilated patients. Median duration of follow-up was 370 days in the ventilated group (range 28 to 6570) and 157 days (range 21 to 5657) in the nonventilated group.

Long-term outcome. A total of 79 (75%) of the ICU patients with GBS had a good outcome (grade ≤ 2) and 26 (25%) had a poor outcome at maximal recovery. Require-

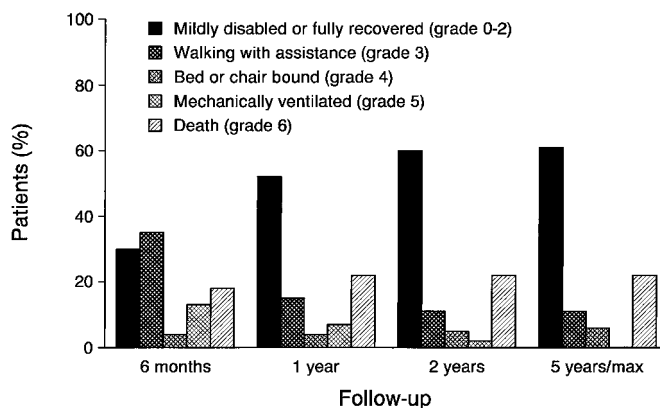


Figure. Detailed analysis of long-term outcome in ventilated patients with Guillain-Barré syndrome.

ment for mechanical ventilation was associated with a poor outcome (21/54 ventilated [39%] versus 5/51 nonventilated [10%], $p < 0.001$) and represented 81% of those with a poor outcome. Detailed analysis of long-term outcome in ventilated patients with GBS is shown in the figure. At 1 year, only 28 patients (52%) had regained independent ambulation, and 4 patients (7%) were still ventilated (grade 5). Recovery of at least one grade beyond 1 year occurred in 10 patients (19%), with 5 patients (9%) improving from poor to good outcome. At maximal recovery, 33 of the ventilated patients (61%) had a good outcome and 21 (39%) had a poor outcome. Mortality related to GBS accounted for over half of the patients with a poor outcome. However, of the 42 survivors, 33 (79%) eventually regained independent ambulation, 6 (14%) could ambulate with assistance (grade 3), and 3 (7%) had severe disability (grade 4) at maximal recovery.

Predictors of maximal recovery outcome. Predictors of outcome at maximal recovery in ventilated GBS patients are presented in table 1. Older age ($p < 0.001$), increased ventilation time ($p = 0.006$), and upper limb paralysis at peak disability ($p = 0.004$) were univariate predictors of a poor maximal outcome. Ventilation longer than 4 months was invariably associated with a poor outcome ($p < 0.001$). Low CMAP amplitudes were not associated with poor outcome. However, the presence of inexcitable nerves on initial electrophysiologic studies was associated with a poor maximal recovery ($p = 0.027$). No significant difference was found with regard to presence of underlying pulmonary disease, preceding GI illness, bulbar dysfunction, bilateral facial palsy, or dysautonomia. Treatment with PE or IVIg in the ventilated group was not associated with outcome. Additionally, although treatment was not routinely administered until 1986, there was no change in outcome over time. Twenty-one of 35 (60%) ventilated patients with GBS between 1976 and 1986 and 12 of 19 (63%) of those between 1987 and 1996 had good outcomes ($p = 1$, Fisher's exact test). Ventilated patients who were admitted directly to Mayo Clinic-affiliated hospitals or transferred within 2 days of presenting to an outside hospital tended to have better outcomes ($p < 0.001$). Using multivariate analysis, increased age (OR 1.99 [for each 10-year increase in age]; 95% CI 1.2 to 3.2, $p = 0.004$) and transfer after 2 days (OR 19.8; 95% CI 3.1 to 128.3, $p = 0.002$) were independent predictors of poor maximal outcome (table 2).

Table 1 Univariate predictors of maximal recovery in ventilated patients with Guillain-Barré syndrome

Clinical feature	Outcome—maximal recovery		<i>p</i> Value*
	Good (n = 33)	Poor (n = 21)	
Age, y (median, range)	53, 11–75	71, 4–87	<0.001
Gender (% male)	61	62	NS
Duration of ventilation, days (median, range)	39, 4–110	70, 3–1825	0.006
Bulbar involvement (%)	97	100	NS
Preceding GI illness (%)	12	14	NS
Upper limb paralysis (%)	3	33	0.004
Median time to peak disability,† days	7	6	NS
Low CMAP amplitude‡ (%)	53	70	NS
Inexcitable nerves§ (%)	3	35	0.005
Treated¶ (%)	58	43	NS
Transferred >2 days (%)	12	62	<0.001

* Two-tailed *p* value from a univariate analysis comparing good vs poor outcome using rank sum test for continuous variables or Fisher's exact test for categorical variables.

† From onset of neuropathic symptoms.

‡ Compound muscle action potential (CMAP) less than 20% lower limit normal (LLN) in at least two nerves.

§ CMAP absent or less than 10% LLN in one nerve and absent in all other nerves tested.¹²

¶ Plasma exchange or iv immunoglobulin, treatment routinely administered 1987–1996.

|| Transferred to Mayo Clinic–affiliated hospitals after spending more than 2 days in outside hospital.

NS = not significant (*p* > 0.05); GI = gastrointestinal.

Predictors of 1-year outcome. To provide comparison with other studies, predictors of 1-year outcome were analyzed. Univariate predictors of poor 1-year outcome in the ventilated patients were older age (*p* < 0.001), duration of ventilation (median 63 versus 37.5 days, *p* = 0.004), upper limb paralysis (*p* = 0.022), inexcitable nerves (*p* = 0.024), delayed time to peak disability (median 10.5 versus 5 days, *p* = 0.044), and delay to transfer (*p* = 0.001). Increased age (*p* = 0.003), prolonged time to peak disability (*p* = 0.045), and delayed transfer (*p* = 0.008) were independent predictors of poor outcome at 1 year (see table 2).

Discussion. Mortality has declined with advances in supportive care, but it remains significant in ventilated patients.² In our series, mortality due to GBS in ventilated patients was 20% and accounted for more than half of those with a poor outcome. The major reasons were ventilator-associated complications in elderly patients with significant associated comorbid illness.⁸ There was no change in mortality over time.⁸ Although the mortality in our series is apparently high, it is comparable to other contemporary observational series in which mortality in ventilated patients ranged from 15 to 30%.^{6,7,14,15} In treatment trials, approximately 5% of patients

Table 2 Multivariate predictors of poor 1-year and poor maximal recovery outcome in ventilated patients with Guillain-Barré syndrome

Clinical feature	Odds ratio		
	Estimate	95% CI	<i>p</i> Value*
One-year recovery			
Age, y†	2.63	1.38–5.03	0.003
Transferred >2 days‡	22.85	2.23–234	0.008
Delayed time to peak disability§	1.11	1.00–1.22	0.045
Maximal recovery			
Age, y†	1.99	1.2–3.2	0.004
Transferred >2 days‡	19.8	3.1–128.3	0.002

* From a multivariate logistic regression model with outcome as the dependent variable and age, time to transfer >2 days, and delayed time to peak disability (1-year outcome only) as independent variables.

† Treated as a continuous variable based on 10-year increments.

‡ Transferred to Mayo Clinic–affiliated hospitals after spending more than 2 days in outside hospital.

§ From onset of neuropathic symptoms; treated as a continuous variable based on 1-day increments.

were excluded before randomization because of significant systemic illness.^{1,9,10,16,17} Direct comparison with these studies may therefore be misleading, as some of the patients who died in our series may have been excluded from these trials on the basis of comorbidity.

In our study, 81% of the patients with a poor outcome required mechanical ventilation. Only 6% of patients with GBS surviving mechanical ventilation were left with severe permanent disability (grade 4). The findings are comparable to one study in which 58 ICU patients (19 ventilated) were followed until maximal clinical recovery. In that series, 5% of patients overall had severe permanent disability (grade 4) at follow-up, and all surviving nonventilated patients eventually regained independent ambulation.³

The same study also reported that weakness may improve for up to 2 years in patients with severe GBS.³ Our results confirm this finding and suggest that improvements may even continue beyond 2 years. Four patients improved from a poor outcome to a good outcome between years 1 and 2, one person regained independent ambulation after 2 years, and one patient could only be completely weaned from nocturnal ventilation after 5 years. This suggests that eventual recovery in patients ventilated for GBS may be prolonged. Treatment trials may underestimate final outcome if follow-up is limited to 1 year.

The selection of a functional score of grade 2 (ability to ambulate independently) as a measure of a good outcome may be debated. Previous studies have generally defined functional grades 0 and 1 as a good outcome but have also used time to independent ambulation (grade ≤2) as a secondary outcome measure.^{6,16,17} At least one large single center study evaluated outcomes in two groups, grades 0 through

2 and grades 3 through 6, and more recently a functional score of 2 or less was utilized to indicate a favorable outcome.^{18,19} Although the Hughes scale is simple, can be administered quickly, and is reliable, it is unable to distinguish between the potentially wide ranges of functional disability within particular grades and has not been systematically validated. The difference between grade 2 and grade 1 is at times difficult to determine,^{7,17,20} particularly in older individuals who are not in the habit of running or performing manual work. For these reasons, we defined a good outcome as grade 0 through 2. The question remains whether this is an adequate marker of functional outcome, and to our knowledge, it has not been compared to other universally accepted outcome scales.

Increased age was highly predictive of poor outcome in our study, which is similar to previous findings.^{6,14} The OR for age on multivariate analysis was 1.99, which indicates that for every 10-year increase in age, the risk for a poor outcome doubles. With regard to other risk factors, in simplest terms, disease severity predicts recovery. Requirement for ventilation is highly correlated with other markers of severe nerve injury and these factors, when present, culminate in ventilation. Duration of ventilation correlated significantly with outcome, which has also been previously noted.^{7,13,18} In our study, ventilation for longer than 4 months was invariably associated with a poor outcome and all patients left with severe permanent disability (grade 4) were ventilated for more than 6 months.

Previous studies have shown that low CMAP amplitude is a predictor of poor outcome.^{8,19,21-23} Our data suggest that the greatest risk of a poor outcome in ventilated patients occurs not in patients with low CMAP amplitudes, but rather in those with inexcitable nerves. The fact that low CMAP amplitude did not predict outcome in ventilated patients may be a reflection of a highly selected group with severe disease, many of whom had evidence of axonal degeneration. Although only two patients with inexcitable nerves had their electrophysiologic studies within 14 days, our findings suggest that the presence of inexcitable nerves at any time during the disease is predictive of a poor outcome. The value of serial nerve conduction studies (NCS) in this subset of patients is not known. However, it has been suggested that inexcitable nerves portend the worst prognosis either early or late in the disease and that NCS at 4 to 6 weeks into the disease are more predictive of outcome.¹² NCS performed early in the disease may not reliably differentiate between conduction block and primary axonal disease but this usually becomes apparent with time.

It has also been suggested that patients with upper limb paralysis (ULP) at peak disability are less likely to ambulate than if any arm movement exists at time of disease nadir.^{16,24} Our study confirms that ULP is a major risk factor for a poor outcome. The reason for this association is not clear. In view of the

correlation with generalized weakness, it most likely reflects the severity of a widespread neuropathic process. Additionally, as all but one patient with ULP required mechanical ventilation, this association may be due to severe involvement of the upper limb nerve roots, which lie in close proximity to those that supply the phrenic nerve. Whereas the scale used in our study is crude, it distinguished paralysis from the presence of any movement. Determination of other degrees of upper limb dysfunction, although important when addressing specific functional abilities, did not add anything to prediction of outcome.

Time to peak disability was predictive of poor 1-year outcome, but not maximal recovery. Our study suggests that in ventilated patients, slow rather than rapid onset to peak disability is predictive of a poor outcome at 1 year. This is in contrast to the majority of studies.^{1-3,22} The exact reason for this is unclear. However, it has been suggested that if intubation is delayed, for whatever reason, atelectasis and aspiration may occur, adversely affecting prognosis due to more prolonged ventilation and a higher rate of ventilator-associated complications.¹⁵ The extent to which this may contribute to poor outcome through mortality is unclear, but consideration of early intubation in older patients may be warranted.

Patients who were seen at this tertiary referral center within 2 days (admitted directly or transferred within 2 days) tended to have better outcomes. This finding is independent of other predictors of poor outcome and does not simply imply that the worst patients were transferred. However, this possibility cannot be completely excluded despite the statistically significant difference. The exact reason for this association is unclear, but may lie in earlier recognition of neuromuscular decline and improved respiratory management. This supports the previous finding of a trend toward improved outcome in patients with GBS who were treated in specialized neurology centers for part or all of their care.¹⁴ We chose transfer within 2 days as it is a reasonable period during which initial assessment and investigation can be completed and to allow for observation for early clinical progression. If this finding is confirmed, it may have significant implications for triage of patients with severe GBS. The increasing use of IVIg in nonspecialized centers in patients who previously might have been transferred for PE may potentially increase future morbidity and mortality.

Mechanically ventilated patients constitute the majority of GBS patients with a poor outcome, and mortality in this population is substantial. However, the majority of survivors regain the ability to ambulate independently and patients may show clinically significant improvement well beyond 1 year. Supportive care therefore remains paramount and increased emphasis should be placed on limiting preventable deaths related to GBS. This may include early referral to a tertiary care center or specialized neurology unit.

Acknowledgment

The authors thank Dr. David J. Plevak, Department of Anesthesia, Mayo Clinic, Rochester, for critical review of the manuscript.

References

1. Plasma Exchange/Sandoglobulin Trial Group. Randomized trial of plasma exchange, intravenous immunoglobulin, and combined treatments in Guillain-Barré syndrome. *Lancet* 1997;349:225–230.
2. Ropper AH, Wijdicks EFM, Truax BT. Guillain-Barré syndrome. Contemporary neurology series. Philadelphia: FA Davis, 1991.
3. Ropper AH. Severe acute Guillain-Barré syndrome. *Neurology* 1986;36:429–432.
4. French Co-Operative Group on Plasma Exchange in Guillain-Barré syndrome. Plasma exchange in Guillain-Barré syndrome: one-year follow-up. *Ann Neurol* 1992;32:94–97.
5. The Italian Guillain-Barré Study Group. The prognosis and main prognostic indicators of Guillain-Barré syndrome. A multicentre prospective study of 297 patients. *Brain* 1996;119:2053–2061.
6. Winer JB, Hughes RAC, Greenwood RJ, Perkin GD, Healy MJR. Prognosis in Guillain-Barré syndrome. *Lancet* 1985;i:1202–1203.
7. Winer JB, Hughes RAC, Osmond C. A prospective study of acute idiopathic neuropathy. 1. Clinical features and their prognostic value. *J Neurol Neurosurg Psychiatry* 1988;51:605–612.
8. Lawn ND, Wijdicks EFM. Fatal Guillain-Barré syndrome. *Neurology* 1999;52:635–638.
9. French Co-Operative Group on Plasma Exchange in Guillain-Barré syndrome. Efficacy of plasma exchange in Guillain-Barré syndrome: role of replacement fluids. *Ann Neurol* 1987;22:753–761.
10. van der Meche FGA, Schmitz PIM, The Dutch Guillain-Barré Study Group. A randomized trial comparing intravenous immunoglobulin and plasma exchange in Guillain-Barré syndrome. *N Engl J Med* 1992;326:1123–1129.
11. Asbury AK, Cornblath DR. Assessment of current diagnostic criteria for Guillain-Barré syndrome. *Ann Neurol* 1990;27(suppl):S21–S24.
12. Hadden RDM, Cornblath DR, Hughes RAC, et al. Electrophysiological classification of Guillain-Barré syndrome: clinical associations and outcome. *Ann Neurol* 1998;44:780–788.
13. Hughes RAC, Newsom-Davis JM, Perkin GD, Pierce JM. Controlled trial of prednisolone in acute polyneuropathy. *Lancet* 1978;ii:750–753.
14. Rees JH, Thompson RD, Smeeton NC, Hughes RAC. Epidemiological study of Guillain-Barré syndrome in southeast England. *J Neurol Neurosurg Psychiatry* 1998;64:74–77.
15. Newton-John H. Prevention of pulmonary complications in severe Guillain-Barré syndrome by early assisted ventilation. *Med J Aust* 1985;142:444–445.
16. Guillain-Barré Syndrome Steroid Trial Group. Double-blind trial of intravenous methylprednisolone in Guillain-Barré syndrome. *Lancet* 1993;341:586–590.
17. The Guillain-Barré Syndrome Study Group. Plasmapheresis and acute Guillain-Barré syndrome. *Neurology* 1985;35:1096–1104.
18. Ng KKP, Howard RS, Fish DR, et al. Management and outcome of severe Guillain-Barré syndrome. *Q J Med* 1995;88:243–250.
19. Rees JH, Soudain SE, Gregson NA, Hughes RAC. *Campylobacter jejuni* infection and Guillain-Barré syndrome. *N Engl J Med* 1995;333:1374–1379.
20. Kleyweg RP, van der Meche FG, Schmitz PI. Interobserver agreement in the strength and functional abilities in Guillain-Barré syndrome. *Muscle Nerve* 1991;14:1103–1109.
21. McKhann GM, Griffin JW, Cornblath DR, Mellits ED, Fisher RS, Quaskey SA. Plasmapheresis and Guillain-Barré syndrome: analysis of prognostic factors and the effect of plasmapheresis. *Ann Neurol* 1988;23:347–353.
22. Emilia-Romagna Study Group on Clinical and Epidemiological Problems in Neurology. Guillain-Barré syndrome variants in Emilia-Romagna, Italy, 1992–93: incidence, clinical features, and prognosis. *J Neurol Neurosurg Psychiatry* 1998;65:218–224.
23. Cornblath DR, Mellits ED, Griffin JW, et al. Motor conduction studies in Guillain-Barré syndrome: description and prognostic value. *Ann Neurol* 1988;23:354–359.
24. Hughes RAC, Hadden RDM, Rees JH, Swan AV. The prognosis and main prognostic indicators of Guillain-Barré syndrome: a multicentre prospective study of 29 patients. *Brain* 1998;121:767–768. Letter.

Neurology®

Long-term outcome in patients with Guillain–Barré syndrome requiring mechanical ventilation

Dade D. Fletcher, Nicholas D. Lawn, Troy D. Wolter, et al.

Neurology 2000;54;2311-2315

DOI 10.1212/WNL.54.12.2311

This information is current as of June 27, 2000

Updated Information & Services

including high resolution figures, can be found at:
<http://www.neurology.org/content/54/12/2311.full.html>

References

This article cites 20 articles, 8 of which you can access for free at:
<http://www.neurology.org/content/54/12/2311.full.html##ref-list-1>

Citations

This article has been cited by 11 HighWire-hosted articles:
<http://www.neurology.org/content/54/12/2311.full.html##otherarticles>

Permissions & Licensing

Information about reproducing this article in parts (figures, tables) or in its entirety can be found online at:
<http://www.neurology.org/misc/about.xhtml#permissions>

Reprints

Information about ordering reprints can be found online:
<http://www.neurology.org/misc/addir.xhtml#reprintsus>

Neurology® is the official journal of the American Academy of Neurology. Published continuously since 1951, it is now a weekly with 48 issues per year. Copyright . All rights reserved. Print ISSN: 0028-3878. Online ISSN: 1526-632X.

